

CWA SCIENCE CLASSES

Concept With Abhishek | Class 11 — Chemistry

Chapter 1: Some Basic Concepts of Chemistry

1. Mole Concept: 1 mole of any substance contains 6.022×10^{23} particles (Avogadro's number).
2. Molar Mass: Mass of one mole of a substance expressed in grams, numerically equal to atomic/molecular mass in amu.
3. Empirical formula shows simplest whole number ratio of atoms; Molecular formula shows actual number of atoms in a molecule.
4. Percentage Composition: Used to determine empirical formula from experimental data.
5. Limiting Reagent: The reactant that gets completely consumed first and limits the amount of product formed.
6. Concentration Terms: Molarity (mol/L), Molality (mol/kg), Mole fraction, Mass percentage, ppm.
7. Laws of Chemical Combination: Law of conservation of mass, law of definite proportions, law of multiple proportions, Gay Lussac's law, Avogadro's law.